

Abstract

India faces an acute informational asymmetry in packaged-food consumption: food labels exist, but food-label literacy does not keep pace with the complexity of modern ultra-processed food formulations. This project report presents **Food Scanner**, a mobile-friendly digital health intervention grounded in the COM-B model of behaviour change. Drawing on scoping reviews, randomized controlled trial evidence, and epidemiological data, the report establishes that functional label literacy in India remains critically low despite growing consumer awareness. Technical lexicons, health-halo marketing, cognitive load, and linguistic exclusion are identified as the primary barriers to meaningful engagement. Food Scanner addresses these barriers by combining optical character recognition (OCR), a rule-based ingredient-flagging engine, and nudge-informed design to translate complex ingredient lists into personalized, plain-language guidance at the point of purchase. The project demonstrates that health psychology — applied through COM-B capability, opportunity, and motivation interventions — can meaningfully bridge the gap between label availability and label usability in the Indian packaged-food context.

1. Introduction

India is undergoing a double burden of disease that reflects both the persistence of undernutrition and the rapid escalation of diet-related non-communicable diseases (DR-NCDs) — obesity, type 2 diabetes (T2D), and cardiovascular disease (CVD). This epidemiological transition is driven substantially by a nutrition transition: a population-level shift from traditional, minimally processed whole-grain diets to a Westernized dietary pattern high in fats, added sugars, and sodium, delivered predominantly through ultra-processed foods (UPFs).

Food labels are the principal regulatory mechanism through which consumers are expected to navigate this transition. Under the 2020 and 2024 regulations of the Food Safety and Standards Authority of India (FSSAI), labels must disclose nutritional content, ingredient lists, allergen information, and additive codes. Yet the presence of a label does not guarantee its utility. Research across thirty-two primary Indian studies confirms that while label awareness is increasing, particularly among urban and educated consumers, **functional literacy — the capacity to access, comprehend, appraise, and act upon label information — remains critically low.** [Pahlani et al., 2025; Suresh et al., 2024]

This project responds to that gap. **Food Scanner** is a mobile-friendly web application that scans or accepts uploaded images of packaged-food labels, extracts ingredient information using OCR, analyses the list against a curated evidence-based database, and returns personalized, plain-language flags tailored to each user's health concerns, allergies, and dietary preferences. The app is grounded in the COM-B model of behaviour change and is informed by nudge theory, health communication research, and the Indian regulatory landscape.

2. Problem Statement

The central problem addressed by Food Scanner is that **food-label availability does not guarantee food-label understanding**. In the Indian context, consumers may be aware of the existence of labels and may even glance at them during purchase, yet still lack the functional literacy needed to interpret ingredient lists, technical additive codes, or front-of-pack claims in health-relevant ways.

2.1 The Awareness–Behaviour Gap

A landmark scoping review by Pahlani et al. (2025) synthesizing thirty-two primary Indian studies found that while awareness of food labels exceeds 80% in most urban samples, actual functional use — defined as reading and acting upon the nutritional facts panel or ingredient list — falls to approximately 30–50%. In Chennai, 70.1% of respondents reported using labels during shopping, yet only 32.1% described label reading as a common practice, and merely 40.2% ever attempted to calculate their caloric intake [Suresh et al., 2024]. The discrepancy between reported awareness and actual behaviour is a robust finding across Indian settings.

2.2 Technical Lexicon as a Barrier

Ingredient lists in Indian packaged foods contain terms — maltodextrin, polysorbate 80, sodium benzoate, acesulfame K — and International Numbering System (INS) codes that are entirely unrecognizable to the average consumer. Technical terms are routinely ignored as jargon [Research Document 1, Table 1], transforming the ingredient list into an exclusive form of expert language rather than a functional public health tool.

2.3 The Health-Halo Effect

Front-of-pack claims such as "low fat," "cholesterol-free," "natural," or "sugar-free" create a **health-halo effect** that misleads consumers into perceiving products as wholesome despite harmful nutritional profiles. A product labelled "zero cholesterol" may legally contain less than 5 mg cholesterol per 100 g while simultaneously containing excessive sodium, saturated fat, or synthetic non-sugar sweeteners [Research Document 1, p. 2]. This dissonance is a primary driver of metabolic dysfunction among unsuspecting consumers.

2.4 Consequences for Public Health

The National CNNS (2019) data reveal that 4.9% of Indian adolescents are already overweight, 10.4% pre-diabetic, and 4.9% pre-hypertensive — trajectories that are directly linked to dietary quality and the consumption of HFSS (High in Fat, Salt, and Sugar) products. As long as the ingredient list remains functionally inaccessible to most consumers, these trends will continue.

3. Need for the App

The reviewed literature establishes five converging reasons why a digital intervention is necessary:

- **Label complexity outpaces consumer capacity.** FSSAI mandates increasingly comprehensive disclosures, yet the cognitive and linguistic demands of reading these labels simultaneously increase.
- **Priority displacement.** Indian consumers prioritize expiry dates (98.5%) and price (92.4%) over nutritional content, largely because safety is immediately observable while chronic-disease risk is diffuse and delayed [Suresh et al., 2024; Kaur & Singh, 2023].
- **INS codes and aliases conceal risk.** Additives linked to gut inflammation, carcinogenicity, and neurobehavioural disruption are listed under codes (e.g., INS 433 for Polysorbate 80) that convey nothing to lay consumers.
- **Health-halo marketing is pervasive and unresisted.** Even consumers with relatively higher nutritional knowledge are susceptible to front-of-pack claims, particularly when purchasing under time pressure.
- **Digital tools reduce cognitive load at scale.** Global evidence, including the FoodSwitch app (George Institute) and experimental studies using traffic-light labelling, demonstrates that digitally mediated interpretation can significantly improve product identification accuracy and, in some contexts, purchase intention.

Food Scanner is therefore needed not as a data repository but as a **literacy translation tool** — one that converts technically opaque ingredient information into understandable, personally relevant, and behaviourally actionable guidance.

4. Literature Review

4.1 Food Label Literacy in India: Determinants and Patterns

Food label literacy in India is shaped by a confluence of socio-demographic, cognitive, and contextual factors. Education is the single strongest predictor: individuals with formal educational backgrounds show

statistically higher knowledge scores ($p = 0.013$), and professional employment status significantly predicts adequate comprehension (AOR: 0.38) [Kaur & Singh, 2023; Suresh et al., 2024]. Gender differences are also documented: female consumers demonstrate consistently higher engagement and awareness than male counterparts. Younger adults (under 40) outperform older cohorts on knowledge assessments ($p < 0.05$), and a marked urban–rural gap persists, with rural adolescents particularly disadvantaged by technical English-language labels [Pahlani et al., 2025].

Determinant	Impact on Literacy	Consumer Response
Socio-demographic (Age, Education)	High	Youth/students show higher awareness but low actual use
Technical Lexicon (INS codes)	Critical Barrier	Terms like INS 433 or Maltodextrin ignored as jargon
FoP Health-Halo Claims	Misleading	Preference for 'Natural' / 'Sugar-Free' without checking NFP
Language / Linguistic Diversity	Moderate–High	Technical English exclusionary for rural consumers
Regulatory Trust Deficit	Mixed	Scepticism toward unverifiable on-pack claims

Even among high-literacy populations, consumption patterns remain driven by heuristics rather than analytical processing. In a study of health-profession students, price and brand name were still prioritized over nutritional panels [Suresh et al., 2024], underscoring that label reading is a **motivated** rather than habitual behaviour — typically triggered only when a consumer has a diagnosed health condition.

4.2 Front-of-Pack Label Formats and Efficacy

FSSAI's FoPL policy debate centres on two competing systems: the **Indian Nutrition Rating (INR)** star system and mandatory **Warning Labels (WL)**. A pivotal randomized controlled trial by Singh et al.

(2022) involving 2,869 adults across six Indian states found Warning Labels (octagonal format) to be the most effective format, enabling 60.8% of participants to correctly identify unhealthy products, compared to 45.0% for Health Star Rating and 39.1% for the control barcode condition. The Supreme Court of India, in early 2026, directed FSSAI to re-evaluate mandatory warning labels, citing the Right to Health as the governing constitutional principle.

FoPL System	Correct Identification (%)	Behavioural Impact
Warning Label (Octagon)	60.8%	Significant deterrent potential
GDA (Numerical)	55.0%	Modest improvement over control
Traffic Light (Colour-coded)	54.8%	Moderate effectiveness
Health Star Rating (HSR)	45.0%	Lowest — susceptible to health halo
Control (Barcode)	39.1%	Baseline — no information

The Star Rating is specifically criticized for the health-halo it creates: an unhealthy product can achieve high ratings by compensating negative nutrients with added fibre or protein, a dynamic exploited by manufacturers.

4.3 Knowledge–Attitude–Practice (KAP) Evidence

Structural Equation Modelling applied to urban Indian residents confirms that knowledge alone does not predict practice. The pathway runs: **Knowledge** → **Attitude** → **Trust** → **Practice**. In South Chennai, 61.7% of consumers held a positive attitude toward food labels, yet only 18.8% expressed full trust in the information provided [Suresh et al., 2024]. This **trust deficit** is a critical bottleneck: when consumers believe manufacturers manipulate labels to appear healthier, they disengage rather than invest cognitive effort in reading.

4.4 Ingredients of Concern: Evidence from the Indian Context

Research Document 1 provides a systematic identification and ranking of food additives clinically relevant to Indian consumers. Six major categories emerge, each combining strong health evidence with high labelling opacity:

Non-Sugar Sweeteners (NSS)

The 2023 WHO guidelines advise against NSS use for weight control following long-term observational data associating aspartame, sucralose, and acesulfame K with increased risk of T2D, CVD, and all-cause mortality. NSS dissociate sweet taste from caloric intake, potentially disrupting cephalic-phase insulin release and gut microbiome composition. Critically, manufacturers conceal these agents behind "no added sugar" claims, rendering them invisible to label-naïve consumers.

Industrial Emulsifiers

Polysorbate 80 (INS 433) and carboxymethylcellulose (CMC, INS 466) — standard UPF texture agents — can alter the intestinal mucus layer, allowing bacterial translocation and inducing low-grade systemic inflammation implicated in metabolic syndrome and inflammatory bowel disease [Naimi et al., 2021]. Their exclusive labelling as INS codes makes them completely opaque to lay consumers.

Synthetic Colorants

FSSAI permits eight artificial food colorants. Even these permitted dyes (e.g., Tartrazine / INS 102; Sunset Yellow / INS 110) are linked to neurobehavioural disruption in children with ADHD. The 2021 OEHHA (California) review concluded that current Acceptable Daily Intake (ADI) thresholds may be set above the levels that trigger neurobehavioural effects in sensitive sub-populations.

Preservatives with Carcinogenic Potential

Sodium nitrite (INS 250), classified by IARC as carcinogenic in processed meats (Group 1), is linked to colorectal cancer through N-nitroso compound formation. Sodium benzoate (INS 211) reacts with ascorbic acid in beverages to form benzene. Cumulative exposure across a UPF-heavy diet remains a public-health concern beyond what any single product's regulatory limit addresses.

Industrial Fats and Refined Carbohydrate Derivatives

Refined palm oil — listed generically as "Vegetable Oil" — elevates LDL-cholesterol via saturated palmitic acid. The ICMR-NIN 2024 guidelines emphasize fat quality and reduced refined carbohydrates as central to Indian dietary recommendations. Maltodextrin, a high-glycaemic saccharide polymer used as thickener and filler, can cause blood-glucose spikes equivalent to table sugar yet escapes "added sugar" designation on nutritional panels — a regulatory loophole of significant public health concern.

Ingredient	Category	Primary Health Concern	Literacy Barrier
Aspartame (INS 951)	Sweetener	T2D, CVD, Metabolic Syndrome	Hidden under 'Sugar-Free' marketing
Polysorbate 80 (INS 433)	Emulsifier	Gut inflammation, MetS	Exclusively listed as INS code
Tartrazine (INS 102)	Colorant	ADHD / hyperactivity (paediatric)	Visual appeal masks neurobehavioural risk
Sodium Nitrite (INS 250)	Preservative	Colorectal cancer (IARC Group 1)	Technical name in processed meats
Maltodextrin	Carbohydrate	Glycaemic load, MetS	Absent from 'added sugar' disclosure
Refined Palm Oil	Fat	LDL elevation, atherosclerosis	Listed generically as 'Vegetable Oil'
Sodium Benzoate (INS 211)	Preservative	Benzene formation, allergy	Ubiquitous in soft drinks
Sucralose (INS 955)	Sweetener	Insulin resistance, gut dysbiosis	Marketed as safer than sugar

4.5 Digital and Behavioural Interventions

The FoodSwitch application (George Institute for Global Health) represents the evidence gold standard: barcode scanning provides real-time Traffic Light or HSR feedback from a database exceeding one million products, enabling point-of-purchase decisions without requiring numeracy or prior nutrition knowledge. Experimental evidence from India using simplified traffic-light food cards in a nudge education paradigm with T2DM adults demonstrated superior HbA1c improvement and dietary compliance over traditional nutritional instruction [MDPI Nutrients, 2025]. Meta-analyses of nudge interventions across settings show an average 15.3% increase in healthy dietary choices, demonstrating that soft behavioural architecture is a low-cost, scalable public health strategy.

5. Theoretical Framework

Food Scanner is primarily structured around the **COM-B Model of Behaviour Change** (Michie et al., 2011), supplemented by nudge theory and KAP-informed health communication principles.

The COM-B Model: Core Proposition

Behaviour (B) occurs when an individual has sufficient Capability (C) to engage in it, the Opportunity (O) to do so, and the Motivation (M) that activates and directs the behaviour. Intervention strategies must address all three components; deficiencies in any one are sufficient to prevent behaviour change.

5.1 Capability

Indian consumers frequently lack both **psychological capability** (the knowledge and comprehension skills to interpret ingredient lists) and, in some contexts, **physical capability** (the literacy or vision to read small-font technical text). Food Scanner addresses both through OCR-mediated text extraction, plain-language ingredient translation, and alias mapping that converts INS codes and brand names of additives into recognizable descriptions of function and risk.

5.2 Opportunity

"Time poverty" in retail environments — the compressed decision window at the point of purchase — eliminates the opportunity for deliberate label analysis even among motivated consumers. Food Scanner restores **physical opportunity** by making information available on the shopper's own device within seconds of scanning. The app also addresses **social opportunity** by designing for eventual multilingual deployment, removing the English-language barrier that currently excludes rural and non-metropolitan consumers from accessing their regulatory rights.

5.3 Motivation

KAP modelling confirms that motivation — not knowledge — is the proximate determinant of label-reading behaviour. Food Scanner enhances **reflective motivation** through personalization: by linking flags to the user's own declared health concerns (e.g., diabetes, PCOS, hypertension, child nutrition), the app makes abstract ingredient risk personally salient. **Automatic motivation** is supported through nudge-aligned design features that reduce future decision friction — saved scans, folders, and favourites function as decision shortcuts that reinforce healthy engagement over time without requiring volitional effort on each occasion.

5.4 Behavioural Nudges

Supplementary nudge mechanisms within Food Scanner include: (a) **salience nudges** — flagged ingredients are visually prominent while safe ingredients are rendered less conspicuous; (b) **framing** — explanations articulate risk in relation to the user's specific health profile rather than abstract population statistics; (c) **default minimization** — reflection prompts are offered but not mandatory, preserving autonomy while scaffolding deliberative processing.

5.5 KAP and Health Communication Principles

The KAP framework cautions that awareness and knowledge are necessary but insufficient conditions for behaviour change. For behaviour to occur, information must be **trusted, understood, and actionable**. Food Scanner's design addresses trust through transparent evidence sourcing; comprehension through plain-language explanation; and actionability through mobile delivery at the moment of relevance.

6. App Aim and Objectives

The overarching aim of Food Scanner is to **improve food-label literacy and support healthier, more reflective packaged-food purchase decisions among Indian consumers.**

The specific objectives are:

1. **Comprehension:** To translate technically opaque ingredient information — including INS codes, chemical aliases, and regulatory euphemisms — into plain-language descriptions that lay consumers can meaningfully interpret.
2. **Personalization:** To personalize ingredient flagging according to each user's self-reported allergies, chronic health conditions, dietary preferences, and sensitivities.
3. **Cognitive load reduction:** To reduce the cognitive burden of label analysis at the point of purchase by automating extraction and prioritizing the most health-relevant information for each individual.
4. **Reflective decision support:** To prompt more deliberate and informed purchase decisions without restricting consumer choice — consistent with COM-B motivation enhancement and nudge theory.
5. **Habit formation support:** To facilitate ongoing, habitual engagement with ingredient information through saved scan histories, custom folders, and favourites that reduce friction on repeat purchases.

7. App Design and Features

Food Scanner is a mobile-first web application. Its architecture and feature set are described below in relation to the COM-B components they serve.

7.1 Onboarding and Profile Setup

Upon sign-in, users configure a personal health profile by selecting from the following preference categories:

- **Allergies:** common allergens including tree nuts, shellfish, dairy, gluten, soy, and egg.
- **Health concerns:** diabetes, hypertension, cardiovascular risk, obesity, PCOS, thyroid conditions.
- **Dietary preferences:** vegetarian, vegan, Jain, halal, kosher, low-FODMAP.
- **Ingredient sensitivities:** NSS, MSG, sulphites, artificial colours, specific preservatives.
- **Custom avoids:** free-text entry for ingredients not covered by default categories.

7.2 Label Scanning and Text Extraction

Users scan or upload images of packaged-food labels. The app applies OCR to extract ingredient text, which is then available for manual correction before analysis — preserving accuracy for degraded or low-contrast label images.

7.3 Ingredient Analysis Engine

Extracted ingredient text is analyzed by a local rule-based engine that:

- maps ingredient names and aliases (including INS codes) to a curated database of 25+ priority additives drawn from the ingredient ranking in Research Document 1;
- assigns confidence levels to flags based on ingredient match precision and evidence strength;
- cross-references the user profile to determine personal relevance and flag priority.

7.4 Results and Explanation Display

Results are displayed as a personalized ingredient flag card. Each flagged ingredient is accompanied by:

- a plain-language description of what the ingredient is and why it was flagged;
- a note on evidence strength (strong / moderate / emerging);
- an explanation of how the ingredient relates specifically to the user's health profile.

7.5 Organisation and Habit Support

Scans are saved to user accounts and can be sorted into custom folders (e.g., Safe Foods, Grocery List, Review Later) or marked as favourites. Personal notes can be appended to any scan. This organizational layer transforms the app from a one-time lookup tool into a developing personal **food knowledge repository**, reducing future decision effort and supporting habit formation consistent with COM-B opportunity and motivation principles.

8. Theory Integration: Why and How

The integration of COM-B and nudge theory into Food Scanner was intentional at the feature level, not merely rhetorical. The table below maps each theoretical construct to the specific design feature that addresses it.

COM-B Component	Barrier Addressed	Feature Response
Psychological Capability	Insufficient ingredient knowledge; INS code opacity	Plain-language flag cards; alias mapping; evidence notes
Physical Capability	Small font; language barriers; low literacy	OCR automation; manual correction; planned multilingual support
Physical Opportunity	Time pressure at point of purchase	Mobile delivery; sub-60-second scan-to-result workflow
Social Opportunity	English-only technical labels exclude rural consumers	Multilingual interface (planned); plain-language default
Reflective Motivation	Lack of personal relevance for generic nutritional info	Profile-personalized flags; health-concern-linked explanations
Automatic Motivation	Habitual brand-trust heuristics override deliberate processing	Saved folders; favourites; reflection prompts; reduced friction

This integration matters because, as the KAP literature demonstrates, consumers do not require **more** information — they require information that is understandable, available at the right moment, and compelling enough to activate behaviour. Food Scanner attempts to satisfy all three conditions simultaneously.

9. Strengths and Limitations

9.1 Strengths

- **Clear public health and literacy problem addressed:** the app responds to a documented, multi-study confirmed gap between label availability and label usability in India.
- **Grounded in health psychology:** the COM-B framework provides a theoretically coherent basis for feature design, ensuring that the app addresses root behavioural determinants rather than surface-level information delivery.
- **Evidence-based ingredient database:** the 25-ingredient priority dataset is derived from peer-reviewed sources with explicit evidence-strength ratings, including IARC, WHO, and ICMR-NIN documentation.
- **Personalization at scale:** the profiling system enables individualized risk communication that is more powerful than population-average nutritional advice for motivating behaviour change.
- **Dual decision horizon:** the app supports both immediate purchase decisions (real-time flags) and longer-term decision strategy (saved scans, folders, favourites).
- **Contextually situated:** the app is designed specifically for Indian regulatory, linguistic, and dietary contexts, rather than adapting a Western tool.

9.2 Limitations

- **OCR accuracy constraints:** image quality, label design variation, and print degradation can reduce extraction accuracy, requiring user correction that adds friction.
- **Database coverage:** the curated dataset of 25 priority ingredients does not yet cover the full breadth of Indian packaged-food additives; novel or obscure ingredients may be missed.
- **Advisory scope:** Food Scanner is an educational decision-support tool, not a diagnostic or clinical instrument; its recommendations cannot replace professional dietetic advice.
- **Absence of formal user evaluation:** no pre/post usability or efficacy study has yet been conducted; the behavioural impact of the app remains to be empirically established.
- **Multilingual gap:** the current version operates primarily in English, limiting reach among the rural and vernacular-language populations most disadvantaged by current labelling norms.
- **Trust calibration:** the app's effectiveness depends partly on users trusting its ingredient assessments; building and sustaining that trust without formal certification remains a challenge.

10. Conclusion

Food Scanner was developed in response to a meaningful, empirically documented gap between food-label availability and food-label usability in the Indian packaged-food market. The literature reviewed across both research documents confirms that Indian consumers — across education levels, genders, and age groups — are constrained in their ability to make label-informed dietary choices by technical complexity, language barriers, health-halo marketing, and cognitive load under real-world retail conditions.

By combining OCR-mediated extraction, a curated evidence-based ingredient database, and a COM-B and nudge-informed design philosophy, Food Scanner translates the inaccessible into the actionable. It does not add to the volume of health information competing for consumer attention; it reorganizes and personalizes existing information to match the capability, opportunity, and motivation profile of the individual user at the moment of decision.

This project demonstrates that health psychology has a substantive and distinctive contribution to make to digital nutrition intervention design — one that extends beyond nutritional science to encompass the cognitive, motivational, and contextual determinants of real-world dietary behaviour. Future development priorities include multilingual expansion, formal usability and efficacy evaluation, and deeper regulatory evidence integration to sustain user trust and population reach.

11. References

The following references underpin the research synthesized in this report. They are organized thematically in the order of their primary relevance to the sections above.

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